MAT 21D Discussion 10 Notes

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§I Stokes' Theorem and Divergence Theorem

(Stokes' Theorem) Let S be an oriented surface with boundary curve C (oriented using the right-hand rule), then for every vector field $\mathbf{F}$ defined everywhere on S, we have

$$\int_C \mathbf{F}\cdot d\mathbf{r} = \iint_S (\nabla\times\mathbf{F})\cdot\mathbf{n}\,d\sigma.$$

(Divergence Theorem) Let D be a solid region with boundary surface S (oriented using the outward normal), then for every vector field $\mathbf{G}$ defined everywhere on D, we have

$$\iint_S \mathbf{G}\cdot\mathbf{n}\,d\sigma = \iiint_D (\nabla\cdot\mathbf{G})\,dV.$$

These two theorems complete the analysis of integrations in three-dimensional space. As I mentioned in the very first discussion, there is a overarching theorem called the **generalized Stokes' theorem** that encompasses all of the analysis we did and generalize them to arbitrary dimensions. It is the reason why I love to share the content of this course with you, and if you are interested, "generalized Stokes' theorem" is the key phrase you can search. You will encounter unfamiliar phrases like "manifolds" and "differential forms," and you can look them up too as if you are peeling an onion. It was a really fun journey for me when I was learning this about "generalized Stokes' theorem," and perhaps, some of you will enjoy it too!

$$\text{Scalar functions } f \xrightarrow{\nabla} \text{Vector fields } \mathbf{F} \xrightarrow{\nabla\times} \text{Vector fields } \mathbf{G} \xrightarrow{\nabla\cdot} \text{Scalar functions } g$$

$$f(B)-f(A) \xleftrightarrow{\text{FT}} \int_C \mathbf{F}\cdot d\mathbf{r} \xleftrightarrow{\text{Stokes' T.}} \iint_S \mathbf{G}\cdot\mathbf{n}\,d\sigma \xleftrightarrow{\text{Div. T.}} \iiint_D g(x,y,z)\,dV$$

$$\text{Points } A,\,B \xleftarrow{\text{End points}} \text{Curves } C \xleftarrow{\text{Boundary}} \text{Surfaces } S \xleftarrow{\text{Boundary}} \text{Solid regions } D$$

Let S be the unit sphere surface (oriented using the outward normal) and let

$$\mathbf{G} = (y\cos 2x)\mathbf{i} + (y^2\sin 2x)\mathbf{j} + (x^2y+z)\mathbf{k}.$$

1. Using the Divergence theorem, compute

$$\iint_S \mathbf{G}\cdot\mathbf{n}\,d\sigma.$$

2. Is there a vector field $\mathbf{F}$ such that $\mathbf{G} = \nabla\times\mathbf{F}$? Explain your answer.

Solution.

1. Divergence theorem tells us that

$$\iint_S \mathbf{G} \cdot \mathbf{n}\, d\sigma = \iiint_D (\nabla \cdot \mathbf{G})\, dV.$$

We can calculate that

$$\nabla \cdot \mathbf{G} = (-2y \sin 2x) + (2y \sin 2x) + 1 = 1.$$

Therefore,

$$\iint_S \mathbf{G} \cdot \mathbf{n}\, d\sigma = \iiint_D 1\, dV = \frac{4}{3}\pi$$

is just the volume of the unit ball D. If we do not know the volume of the unit ball, we can also calculate it using spherical coordinates, so

$$\iiint_D 1\, dV = \int_0^{2\pi} \int_0^{\pi} \int_0^{1} \rho^2 \sin(\varphi)\, d\rho\, d\varphi\, d\theta = \frac{4}{3}\pi.$$

2. Note that for any vector field $\mathbf{F}$, we have

$$\nabla \cdot (\nabla \times \mathbf{F}) = 0.$$

We shown this in homework 8. However, because $\nabla \cdot \mathbf{G} \neq 0$, we know that there does not exist a vector field $\mathbf{F}$ such that $\mathbf{G} = \nabla \times \mathbf{F}$.

Alternatively, we can argue with Stokes' theorem. If $\mathbf{G} = \nabla \times \mathbf{F}$, then we must have

$$\iint_S \mathbf{G} \cdot \mathbf{n}\, d\sigma = \iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n}\, d\sigma = \int_C \mathbf{F} \cdot d\mathbf{r}$$

where C is the boundary curve of S. In particular, for the unit sphere surface S which has no boundary curve, the line integral is zero, so we must have

$$\iint_S \mathbf{G} \cdot \mathbf{n}\, d\sigma = 0.$$

But we have shown in the previous part that this is not the case, so there does not exist a vector field $\mathbf{F}$ such that $\mathbf{G} = \nabla \times \mathbf{F}$.

§II Final Review

The tables in this section are checklists of skills you are expected to learn. Rate your grasp of each skill from

- 1: I don't know how to do this at all.
- 5: I am confident that I can do this on my own without looking at any notes.

After each checklist, there are exercises for you to practice selected by me personally. The exercises are all pulled from discussions, homeworks, and exams, so you have the solutions to them.

Double integral in rectangular coordinates (Sections 15.1-15.3)

	1	2	3	4	5
Describe a region given $a \le x \le b$ and $g_1(x) \le y \le g_2(x)$ (or given $c \le y \le d$ and $h_1(x) \le x \le h_2(x)$)					

	1	2	3	4	5
Given a description of a region, set up an iterated integral of a function $f(x,y)$ over this region					
Exchange the order of integration in an iterated integral while taking care of the bounds					
Calculate the area of a region using rectangular coordinates					
Know how to deal with piecewise behavior					

Exercise 1: Midterm 1A, question 1

Let E denote the region in the plane bounded between the line $y = 0$, the line $2y + x = 4$, and the parabola $x = y^2 - 4$.

(a) Sketch the region.

(b) Evaluate the double integral

$$\iint_E y\, dA$$

as an iterated integral where the integration is performed in the order $dx\, dy$

Exercise 2: Homework 1, question 3

Sketch the region of integration for the integral

$$\int_0^1 \int_0^{x^2-2x+1} xy\, dy\, dx.$$

and write an equivalent integral with the order of integration reversed.

Exercise 3: Discussion 1

Consider the region R bounded by $y = x^2$, $x = 0$, $y = 1$ in the first quadrant. Write down the two iterated integrals in rectangular coordinates that

$$\iint_R xe^{y^2}\, dy\, dx$$

is equal to, and evaluate either one of them.

Exercise 4: Midterm 1B, question 1

Let E denote the triangular region in the plane with vertices at $(0, 0)$, $(2, 0)$, and $(4, 4)$.

(a) Evaluate the double integral

$$\iint_E xy\, dA$$

as an iterated integral where the integration is performed in the order $dx\, dy$.

(b) If the integration is performed in the reverse order $dy\, dx$, the double integral needs to be broken up into a sum of two iterated integrals for the evaluation to be carried out. Write down these two iterated integrals.

Double integral in polar coordinates (Section 15.4)

	1	2	3	4	5
Understand the curve described by $\theta = \alpha$ (for some constant α)					
Understand the curve described by $r = g(\theta)$					
Describe a region given $\alpha \leq \theta \leq \beta$ and $g_1(\theta) \leq r \leq g_2(\theta)$					
Given a description of a region (possibly in rectangular coordinates), set up an iterated integral of a function $f(r, \theta)$ over this region					
Calculate a double integral using polar coordinates					

Exercise 5: Practice midterm 1, question 1

The integral

$$\int_0^\pi \int_0^3 (3 - r) r\, dr\, d\theta$$

is the representation of the double integral

$$\iint_R f(x, y)\, dA$$

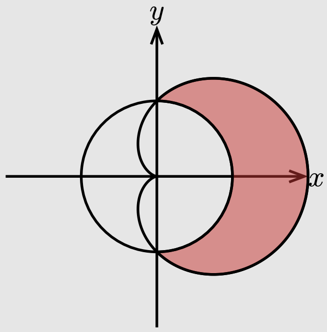
for a suitable region R and a suitable function $f(x, y)$.

(a) Write down a formula for $f(x, y)$ and sketch the region R.

(b) Describe the solid region in three-dimensional space whose volume is calculated by this integral.

Exercise 6: Discussion 2

The figure below shows two curves: the circle $r = 1$ and the cardioid $r = 1 + \cos(\theta)$. Consider the shaded region R in the figure.



Calculate the area of the region R using polar coordinates.

Exercise 7: Midterm 1B, question 1

Let E denote the triangular region in the plane with vertices at $(0, 0)$, $(2, 0)$, and $(4, 4)$. Rewrite the double integral

$$\iint_E xy\, dA$$

as an iterated integral in polar coordinates.

Exercise 8: Midterm 1A, question 3

Evaluate the iterated integral

$$\int_0^1 \int_0^{\sqrt{1-x^2}} (x^2 + y^2)\, dy\, dx$$

by converting it to an iterated integral in polar coordinates.

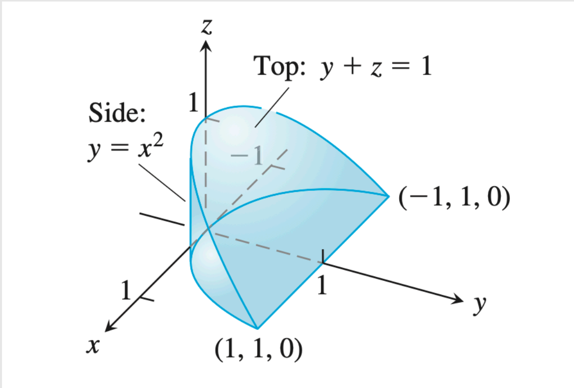
Triple integral in rectangular coordinates (Section 15.5)

	1	2	3	4	5
Given a solid region, set up an iterated integral of a function $f(x, y, z)$ over this region					
Calculate a triple integral of a region using triple integrals (not on the final exam)					
Exchange the order of integration in a triple integral while taking care of the bounds					

Exercise 9: Discussion 3

Here is the region of integration of the integral

$$\int_{-1}^{1}\int_{x^2}^{1}\int_{0}^{1-y} dz\,dy\,dx.$$



Rewrite the integral as an equivalent iterated integral in the order

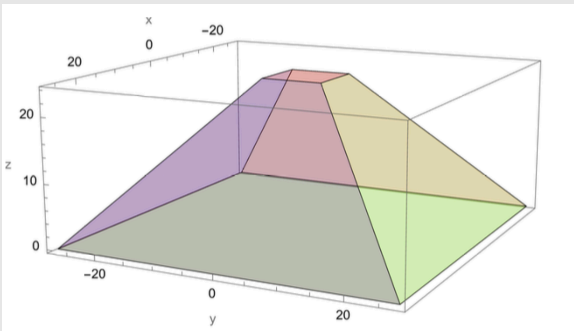
(a) $dz\,dx\,dy$,

(b) $dy\,dz\,dx$,

Exercise 10: Midterm 1B, question 2

Consider the following truncated square pyramid (centered symmetrically around the z-axis) with the following dimensions:

- Width of base $B = 54$, the xy-cross-section is the square $-27 \le x \le -27$ and $-27 \le y \le 27$.
- Width of top $T = 10$, the xy-cross-section is the square $-5 \le x \le 5$ and $-5 \le y \le 5$.
- Height $H = 24$.



(a) The xy-cross-section at half of height $z = 12$ is a square with width $\frac{54}{2} + \frac{10}{2} = 32$. Describe this xy-cross-section as $a \le x \le b$ and $c \le y \le d$.

(b) In general, the xy-cross-section at the height z is a square with width $(1 - \frac{z}{24})54 + (\frac{z}{24})10$. Describe the xy-cross-section at arbitrary height z as $a(z) \le x \le b(z)$ and $c(z) \le y \le d(z)$.

(c) Write down an iterated integral in the order $dx\,dy\,dz$ for the the volume of the truncated square pyramid.

Triple integral in cylindrical and spherical coordinates (Section 15.7)

	1	2	3	4	5
Understand the surface described by $\theta = \alpha$ (for some constant α)					
Understand the surface described by $r = a$ (for some constant a)					
Understand the surface described by $z = g(\theta, r)$ (for some function g)					
Understand the surface described by $\varphi = \alpha$ (for some constant α)					
Understand the surface described by $\rho = g(\theta, \varphi)$					
Set up an iterated integral of a function $f(r, \theta, z)$ over a solid region in cylindrical coordinates					
Set up an iterated integral of a function $f(\rho, \theta, \varphi)$ over a solid region in spherical coordinates					

Exercise 11: Midterm 1B, question 3

(a) Find an equation in spherical coordinates for the cone $z = \sqrt{x^2 + y^2}$.
(b) Find an equation in rectangular coordinates for the surface $\theta = \frac{\pi}{4}$.
(c) Describe in plain language the solid region $0 \leq \theta \leq 2\pi$, $0 \leq \varphi \leq \frac{\pi}{3}$, and $0 \leq \rho \leq 2$.

Exercise 12: Homework 4, question 1

Consider the integrated integral in cylindrical coordinates

$$\int_0^{2\pi} \int_0^{\frac{1}{2}} \int_{-\sqrt{1-r^2}}^{\sqrt{1-r^2}} f(r, \theta, z) r \, dz \, dr \, d\theta.$$

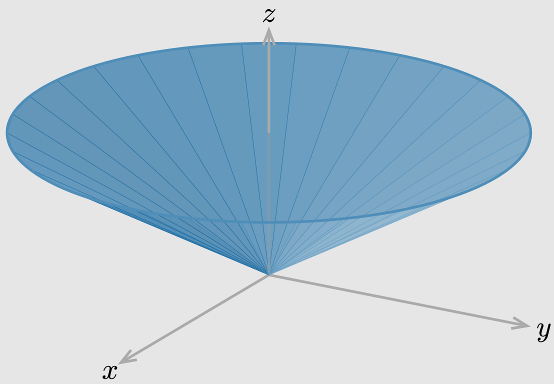
(a) Describe/sketch the region of integration.
(b) Convert the integral to rectangular coordinates in the the order $dz \, dy \, dx$.

Exercise 13: Discussion 4

(a) Describe the region of integration for the integral

$$\int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{2}} \int_0^{2} f(\rho, \theta, \varphi)\rho^2 \sin(\varphi)\, d\rho\, d\varphi\, d\theta.$$

(b) Express as a triple integral in spherical coordinates the volume of the cone obtained by connecting the boundary of the disk described by the two equations $x^2 + y^2 \leq 12$ and $z = 2$ to the origin.



Substitutions in multiple integrals (Section 15.8)

	1	2	3	4	5
Transform a region using inequalities					
Transform a region using boundary curves					
Understand why $dA = r\, dr\, d\theta$ using substitutions					
Understand why $dV = r\, dz\, dr\, d\theta$ using substitutions					
Understand why $dV = \rho^2 \sin(\varphi)\, d\rho\, d\varphi\, d\theta$ using substitutions					
Perform substitutions with x and y as functions of u and v: $$\iint_R f(x, y)\, dx\, dy = \iint_G F(u, v) \left\lvert \frac{\partial(x, y)}{\partial(u, v)} \right\rvert du\, dv$$					
Perform substitutions with u and v as functions of x and y: $$\iint_R f(x, y) \left\lvert \frac{\partial(u, v)}{\partial(x, y)} \right\rvert dx\, dy = \iint_G F(u, v)\, du\, dv$$					
Perform substitutions in triple integrals					

Exercise 14: Discussion 5

Let R be a tilted square in the xy-plane with vertices $(1,0)$, $(0,1)$, $(-1,0)$, and $(0,-1)$.

(a) Region R is bounded by 4 lines. Reexpress R using inequalities based on the equations of these lines.

(b) Consider the transformation

$$x = \frac{u+v}{2}, \quad y = \frac{u-v}{2}.$$

Find the inverse transformation and use it to find the region G in the uv-plane that corresponds to R.

(c) Evaluate the double integral

$$\iint_R (x+y)^2 e^{x-y}\, dx\, dy.$$

Exercise 15: Discussion 5

Let R be the region in the first quadrant of xy-plane bounded by the curves

$$xy = 1, \quad xy = 4, \quad y = x, \quad y = 2x.$$

Evaluate the double integral

$$\iint_R \frac{y}{x}\, dx\, dy$$

using the substitution $u = xy$ and $v = \frac{y}{x}$.

Exercise 16: Homework 5, question 3

(a) Compute the Jacobian of the map

$$\begin{cases} x = u \\ y = u + v \\ z = u + v + w \end{cases}$$

(b) Use the Jacobian calculation to convert the triple integral

$$\int_0^1 \int_x^{x+2} \int_y^{y+3} \left(y^2 - x^2\right) dz\, dy\, dx$$

to an equivalent integral in uvw-coordinates.

Exercise 17: Practice midterm 2, question 1

Consider the transformation

$$\begin{cases} x = e^u \cos v, \\ y = e^u \sin v. \end{cases}$$

Let R be the standard unit square in the uv-plane defined by $0 \le u \le 1$ and $0 \le v \le 1$. Write down the equations of the boundary curves of S in the xy-plane that corresponds to R under this transformation and sketch the region S.

Geometry of curves (Sections 13.1-13.5)

	1	2	3	4	5
Find arc length parameterization					
Formula will be given for quantities of curves, just need to know how to take derivatives and magnitudes					
Know which quantities are independent of parameterization					
Parameterize a curve given description					

Exercise 18: Midterm 2B, question 2

Let curve C be given by the parametric equations

$$\mathbf{r}(t) = (3\cos t)\mathbf{i} + (3\sin t)\mathbf{j} + 4t\mathbf{k}, \quad 0 \le t \le 8\pi.$$

(a) Find the arc length parameter $s(t) = \int_0^t |\mathbf{r}'(t')|\, dt'$.
(b) Write down a reparameterization $\mathbf{q}(s)$ in terms of the arc length parameter.
(c) Compute the tangent vector $\mathbf{T} = \frac{d\mathbf{r}/dt}{|d\mathbf{r}/dt|}$.
(d) Compute the curvature $\kappa = \frac{1}{|d\mathbf{r}/dt|}\left|\frac{d\mathbf{T}}{dt}\right|$.
(e) Compute the principal unit normal $\mathbf{N} = \frac{d\mathbf{T}/dt}{|d\mathbf{T}/dt|}$.

Also see practice midterm 2, midterm 2A, homework 6, and discussion 6 for more exercises.

Between potential functions and line integrals (Section 16.1-16.3)

	1	2	3	4	5
Compute line integrals of scalar functions					
Compute line integrals of vector fields					
Use fundamental theorem of calculus for line integrals					
Determine if a vector field is conservative using component test					
Find a potential function for a conservative vector field					
Decompose a vector field into conservative and non-conservative parts					

Exercise 19: Midterm 2B, question 1

Consider the vector fields

$$\mathbf{F}(x,y) = 2y\mathbf{i} - 2y\mathbf{j} \quad \text{and} \quad \mathbf{G}(x,y) = (x^2+y^2)\mathbf{i} + 2xy\mathbf{j}.$$

Let C be the curve with the parameterization

$$\mathbf{r}(t) = t\mathbf{i} + t^2\mathbf{j}, \quad 0 \le t \le 1.$$

(a) Explain whether there exist a potential function $U(x,y)$ such that $\mathbf{F} = \nabla U$? If so, find one.
(b) Explain whether there exist a potential function $V(x,y)$ such that $\mathbf{G} = \nabla V$? If so, find one.
(c) Evaluate

$$\int_C \mathbf{F} \cdot d\mathbf{r}.$$

Use the fundamental theorem of calculus for line integrals if possible.
(d) Evaluate

$$\int_C \mathbf{G} \cdot d\mathbf{r}.$$

Use the fundamental theorem of calculus for line integrals if possible.

Exercise 20: Discussion 7

Determine if the vector field

$$\mathbf{F}(x,y,z) = e^y\mathbf{i} + (xe^y + z)\mathbf{j} + y\mathbf{k}$$

is conservative. If it is, find a potential function $U(x,y,z)$ that satisfies $U(0,0,0) = 1$.

Exercise 21: Discussion 8

Let $\mathbf{F}(x,y) = (e^x \cos y)\mathbf{i} + (x - e^x \sin y)\mathbf{j}$ and C be the line segment from $(0,0)$ to $(1,1)$. Evaluate the line integral

$$\int_C \mathbf{F} \cdot d\mathbf{r}$$

by decomposing $\mathbf{F}$ into a conservative part and a non-conservative part.

Between line integrals and surface integrals (Sections 16.4-16.7)

	1	2	3	4	5
Compute surface integrals of scalar functions					
Compute surface integrals of vector fields					
Use Green's or Stokes' theorem to evaluate integrals					
Use Green's theorem to compute area					

Exercise 22: Discussion 9

Let S be the portion of the plane

$$x + y + z = 4$$

that lies above the square $0 \leq x \leq 1$, $0 \leq y \leq 1$ in the xy-plane.

(a) Draw a picture of the surface S.

(b) Rewrite the surface integral

$$\iint_S z \, d\sigma$$

as an iterated integral in $du\,dv$.

Exercise 23: Discussion 9

Rewrite the surface integral

$$\iint_S (x - y - z) \, d\sigma$$

as iterated integrals for each of the following surfaces S.

(a) The portion of the plane $x + y = 1$ in the first octant between $z = 0$ and $z = 1$.

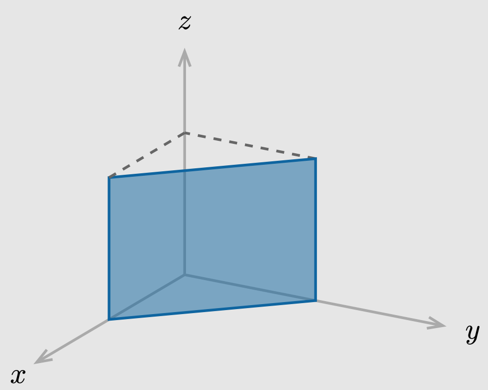


Figure 6: Picture of the portion of the plane $x + y = 1$.

(b) The portion of the sphere $x^2 + y^2 + z^2 = 1^2$ in the first octant.

Exercise 24: Homework 10, question 1

The torus is a surface given in parametric form by

$$\mathbf{r}(u, v) = \cos(u)(r\cos(v) + R)\mathbf{i} + \sin(u)(r\cos(v) + R)\mathbf{j} + r\sin(v)\mathbf{k}$$

where R is the major radius and r is the minor radius, and both parameters u and v range from 0 to 2π. Calculate the surface area of the torus.

Exercise 25: Homework 10, question 2

Calculate the surface integral $\iint_S \mathbf{F} \cdot \mathbf{n} \, d\sigma$ where $\mathbf{F} = x^2\mathbf{i} + \mathbf{j} + xz\mathbf{k}$ and S is the surface given in parametric form by

$$\mathbf{r}(u, v) = v\cos(u)\mathbf{i} + v\cos(u)\mathbf{j} + v\sin(u)\mathbf{k}$$

for $0 \leq u \leq 1$ and $0 \leq v \leq 2\pi$.

There is also homework 9 for a question about Green's theorem. Stokes' theorem is potentially on the final exam, but the question will be more conceptual it seems based on professor Romik's announcement. There is no homework coverage for Stokes' theorem, so I doubt the question would be computational.

Between surface integrals and triple integrals (Section 16.8)

	1	2	3	4	5
Use divergence to determine whether a vector field is a curl of a vector field					
Use divergence theorem to write down equivalent integrals					

This very discussion! But divergence theorem would not be on the final exam..